

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

TEAM ID	SWTID2026-3387
PROJECT TITLE	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
MAX MARKS	5 MARKS

CHAPTER – VII

FUNCTIONAL AND PERFORMANCE TESTING

7.1 – Performance Testing & Functional Testing

Performance Testing

1. Dashboard Loading Speed

- Verified that all Tableau dashboards load within an acceptable time.
- Ensured users can access visualizations without unnecessary delays.

2. Data Processing Performance

- Tested the dashboard using the complete housing dataset.
- Confirmed that all records were processed efficiently without missing data.

3. Filter Response Time

- Checked the response time of filters such as house age, renovation year, and sale price.
- Ensured filtered results were displayed instantly.

4. Chart Rendering

- Verified that bar charts, pie charts, line charts, and histograms rendered correctly.
- Ensured visualizations loaded without graphical errors.

5. Dashboard Navigation

- Tested navigation between dashboards and story pages.
- Confirmed smooth transitions without delays.

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6. User Interaction

- Evaluated interactions such as selecting marks, hovering over charts, and drill-down analysis.
- Verified that all interactive features responded correctly.

7. Large Dataset Performance

- Tested dashboard performance with thousands of housing records.
- Ensured the dashboard remained responsive during analysis.

8. System Stability

- Performed continuous usage testing for an extended period.
- Confirmed there were no crashes, freezes, or unexpected errors.

9. Browser Compatibility

- Tested the dashboard in multiple web browsers.
- Verified consistent performance and display across supported browsers.

10. Resource Utilization

- Monitored CPU and memory usage during dashboard execution.
- Ensured efficient utilization of system resources.

11. Visualization Refresh

- Verified that all charts refreshed automatically after filter selections.
- Confirmed accurate updates without requiring manual reloads.

12. Overall Performance Evaluation

- The Tableau dashboards demonstrated excellent speed, responsiveness, and stability.
- Users could perform housing market analysis efficiently with minimal waiting time.

Functional Testing

1. Dashboard Accessibility

- Verified that every dashboard opened successfully without errors.

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- Confirmed users could access all project pages.

2. Data Accuracy

- Compared dashboard values with the original housing dataset.
- Verified that all displayed information was accurate.

3. Average Sale Price Calculation

- Tested the calculated field for average sale price.
- Confirmed that calculations matched the dataset values.

4. Basement Area Analysis

- Verified that basement area values were displayed correctly.
- Confirmed charts represented the correct measurements.

5. House Age Distribution

- Tested the house age visualization for correctness.
- Verified accurate grouping and distribution of house ages.

6. Renovation Analysis

- Checked renovation year calculations and visualizations.
- Confirmed that renovated and non-renovated houses were classified correctly.

7. Filter Functionality

- Tested all dashboard filters individually and together.
- Ensured that every filter updated related charts correctly.

8. Tooltip Verification

- Verified that tooltips displayed complete and accurate information.
- Ensured users received meaningful details when hovering over charts.

9. Story Navigation

- Tested navigation between all Tableau story points.
- Confirmed that every story page loaded correctly.

10. Chart Validation

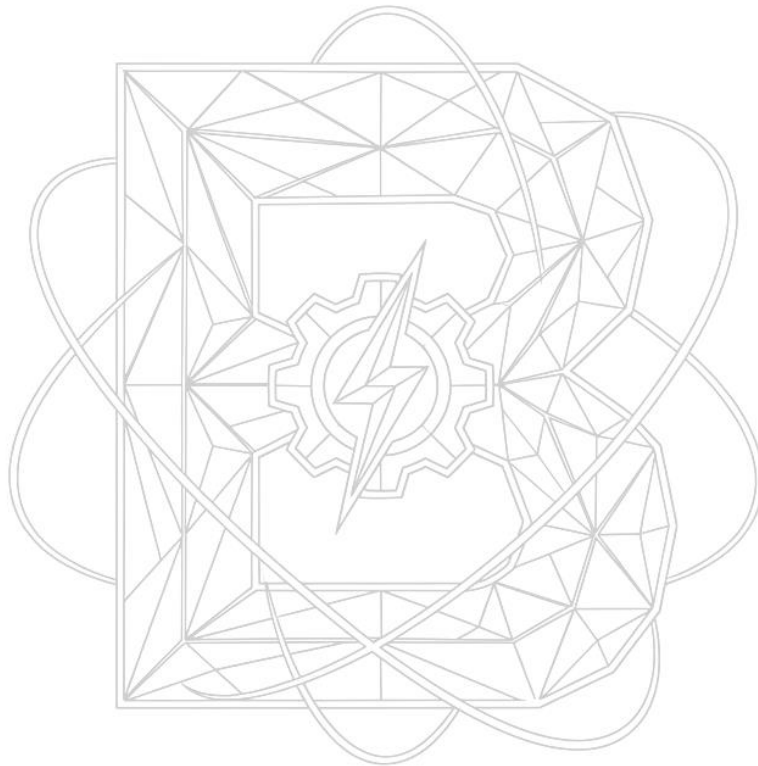
- Verified that all charts displayed the correct labels, legends, titles, and axes.

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- Confirmed visual consistency throughout the dashboard.

11. User Interaction Testing

- Tested selecting, highlighting, filtering, and drill-down features.
- Ensured all interactive functions operated smoothly.



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